

A Literature Roadmap for VoxSight Recall

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Purpose: Build a rigorous mental model of **episodic visual memory**, from its historical roots to contemporary long-context multimodal systems, and identify the literature most relevant to a publishable VoxSight Recall project.

How to use this roadmap

Do not begin by treating this as a single field called “large-language visual memory.” It is a **convergence of six research lineages**: human episodic memory, personal external memory, neural and retrieval-based memory, long-video multimodal models, egocentric-video retrieval, and assistive wearable systems. The main intellectual task is to learn what each lineage solves—and, just as importantly, what it does **not** solve.

The working definition for your project: VoxSight Recall is a system that creates and retrieves **time-stamped, evidence-linked visual event memories** from a user’s past camera observations, then communicates the result with calibrated uncertainty and accessible spoken guidance. It is not a claim that a machine possesses human consciousness or human episodic recollection.

The accompanying `vox sight_literature_mental_map.png` is the high-level map. Read the document from top to bottom once, then follow the four-week reading sequence near the end.

1. The one-page mental model

Layer	Core question	Canonical idea	What it contributes to VoxSight Recall	What it does not solve
Human episodic memory	What counts as a memory of an event?	An episode binds an event to time and place.	Defines why “I last saw your keys on the kitchen counter at 10:18” is more than a fact.	It does not provide a computer architecture.
Personal external memory	Why capture a personal record at all?	Memex, lifelogging, and wearable memory aids.	Establishes the human-centered vision of searchable	It does not provide moderate visual semantic search or safe L responses.

			personal experience.	
External AI memory	Where should the memory live?	External stores accessed by retrieval, not only model weights.	Supports a database of editable, auditable visual events.	It does not solve visual grounding or temporal state change.
Long-video MLLMs	How can a model handle long visual histories?	Compress frames and retain a bounded memory bank.	Provides efficient online processing mechanisms.	It may discard too much exact evidence that a user needs.
Egocentric episodic retrieval	How is past first-person video queried and evaluated?	Localize the relevant moment in past video.	Provides datasets, tasks, and objective retrieval metrics.	It often stops at past-event localization rather than actionable current help.
Assistive visual memory	Why does this matter to users?	Wearable, voice-mediated object retrieval.	Validates the real need for “where did I put X?” interactions.	It leaves room for stronger provenance, uncertainty, privacy, and accessibility research.

The essential chain is:

Camera observation → select important event → represent time/object/action/place/evidence → store → retrieve under a spoken query → verify and calibrate → communicate an actionable answer.

Every paper you read should be placed at one point in this chain. A long-video paper may improve **selection and representation**. A RAG paper may improve **retrieval and grounding**. A user-study paper may improve **communication and usefulness**. Your research contribution should be the connective tissue that makes the whole chain trustworthy for a blind or low-vision user.

2. Where the idea started: memory before machine learning

2.1 The Memex: a personal memory system before computers were personal

Vannevar Bush's 1945 essay *As We May Think* proposed the **Memex**, a personal device for storing and traversing large collections of records through associative trails. ¹ The device was not a camera assistant, and it did not contain artificial intelligence. Its enduring contribution is conceptual: personal knowledge systems should help a person **return to a meaningful prior item** rather than merely archive data.

For VoxSight, Bush gives you the oldest framing of the problem: people do not want more captured video; they want a reliable path back to a relevant experience. Modern retrieval systems replace associative trails with embeddings and language queries, but the user value remains the same.

2.2 Episodic versus semantic memory

Endel Tulving's distinction between **episodic** and **semantic** memory is the conceptual anchor for your terminology. ² Episodic memory concerns events tied to their temporal and spatial context; semantic memory concerns general facts and concepts. A user's "keys are usually near the entrance" is semantic-like knowledge. "At 10:18, the user placed the keys on the counter beside the fruit bowl" is an episodic event record.

Later cognitive work emphasizes that these systems are distinguishable but interact during encoding and retrieval. ³ That matters technically. A visual assistant needs semantic labels such as *keys*, *counter*, and *kitchen*, but it must not replace event evidence with what is merely plausible. The correct response is not "your keys are probably near the door" ; it is "I last observed them on the kitchen counter at 10:18. I cannot verify whether they are still there."

Conceptual discipline: Use "episodic visual memory" as the motivating term, but use **evidence-grounded event memory** as the technical definition in a paper. This avoids overstating an analogy between a stored machine record and human subjective recollection. ⁴

2.3 Lifelogging and wearable memory aids

The route from Memex to VoxSight passes through **lifelogging**. MyLifeBits explored a personal digital repository intended to retain a lifetime of documents, media, and other records. ⁵ It separates three problems that should not be conflated: **capture**, **storage**, and **retrieval**.

SenseCam made the idea wearable. Hodges et al. described a sensor-augmented camera that automatically captures a visual record of a wearer's day and was designed as a retrospective memory aid. ⁶ SenseCam is historically important because it shows that

wearable visual memory began as a human-centered assistive technology long before today’s multimodal language models.

Historical question	Early answer	Modern VoxSight extension
Can a personal life record be captured?	Lifelogging and SenseCam.	Yes, but capture should be selective and privacy-aware.
Can it be searched?	Manual review, metadata, conventional databases.	Voice queries and multimodal retrieval.
Can it understand a query such as “Where did I put my keys?”	Only weakly.	Parse the query into object, action, time, place, and desired confidence.
Can it safely answer the user?	Not the central focus.	Distinguish last observed from currently present , cite evidence, and speak uncertainty clearly.

3. How memory entered modern AI

3.1 Differentiable external memory

Neural Turing Machines (NTMs) introduced a learnable controller coupled to an external memory that can be read and written through attention. ⁷ This is a foundational neural architecture result, not a direct product blueprint. Its lesson is that **memory can be external to the core model** rather than permanently encoded in model parameters.

For a solo project, do not begin by training an NTM or a differentiable memory model. Its value is conceptual: it teaches you to think in terms of memory **operations**—write, retrieve, update, and delete—rather than viewing an LLM prompt as the only place memory exists.

3.2 Retrieval-augmented generation: the key production pattern

Retrieval-Augmented Generation (RAG) combined a parameterized generator with a non-parametric external index accessed by a retriever. ⁸ The original paper explicitly noted limitations in updating knowledge and providing provenance. This makes RAG central to VoxSight: a visual-memory system should retrieve explicit past evidence before generating a spoken response.

The critical adaptation is that your retrievable unit cannot be an ordinary text chunk. It should be a **visual event record**. A minimal record might contain:

Field	Example	Why it matters
event_id	evt_2026_07_18_1018	Auditable unit of retrieval.
time_interval	10:18:04–10:18:09	Lets the assistant state when it observed the event.
object	keyring	Main retrieval target.
action/state	placed on counter	Distinguishes placement from mere visibility.
place cues	kitchen; counter; beside fruit bowl	Gives action-relevant spatial language.
visual evidence	keyframe ID, crops, embeddings	Allows verification and later review.
confidence	calibrated score and reason	Prevents a confident but unsupported answer.
privacy policy	retention window, deletion flag	Makes the memory controllable.

3.3 LLM agent memory

Generative Agents demonstrated an architecture that records observations, synthesizes higher-level reflections, and retrieves memories dynamically for planning. ⁹ MemGPT explored tiered memory management inspired by operating systems, separating limited in-context memory from larger external memory. ¹⁰ MemoryBank studied long-term conversational memory with update and forgetting mechanisms. ¹¹

These works answer a different question from visual memory: **how can an LLM manage extended interaction histories?** They are useful for memory policy but insufficient for VoxSight because textual summaries alone can lose object identity, fine-grained spatial relations, and proof. Your system needs a hybrid: event text for language retrieval plus links to visual embeddings and keyframes for grounding.

4. How large multimodal models remember long videos

4.1 The scale problem

A video is not a short prompt. It contains a continuous flow of images, and a system that sends every frame to an MLLM will quickly exceed context and GPU limits. Long-video

memory papers therefore focus on **selection and compression**. Their central technical question is not “what did a person last do with their keys?” but “how can a model retain useful information across a long video within bounded compute?”

4.2 MovieChat: sparse memory from dense video tokens

MovieChat treats tokens as carriers of memory and uses a short-/long-term structure to compress dense video information into sparse memory representations for long-video understanding.¹² Its contribution is the general principle that persistent visual memory must be selective. The caution for VoxSight is equally important: a generic compression procedure may discard the tiny frame containing the key placement, which is exactly the event that later matters.

4.3 MA-LMM: online processing with a memory bank

MA-LMM processes video online and stores earlier visual information in a long-term memory bank, allowing later reference without loading the full video into a multimodal model context.¹³ It is the closest architectural neighbor to the idea of a continuous visual assistant. The paper supports the idea that your system should not replay all history on every user query.

However, MA-LMM is primarily a **model-level long-video understanding** solution. It does not by itself give you a user-facing event schema, state-change reasoning, calibrated last-seen language, privacy controls, or a procedure for helping a user physically act on the answer.

Long-video question	MovieChat / MA-LMM answer	VoxSight-specific research question
What should be retained?	Compressed feature/tokens useful for tasks.	Preserve high-value event evidence such as object placement and relocation.
How is history accessed?	Memory bank / sparse token mechanism.	Query-conditioned retrieval over event records with visual verification.
What counts as success?	Video QA/captioning/understanding score.	Correct event, usable location cue, truthful confidence, and successful user action.
How is current state handled?	Often not the target.	“Last observed” must not become “currently there.”

5. Egocentric video turns visual history into a research task

5.1 Ego4D episodic memory

Ego4D established a large-scale first-person-video research ecosystem. Its episodic-memory benchmark frames the problem as querying the user's own past video and localizing where the relevant answer is visible.¹⁴ The benchmark includes natural-language queries, visual queries, and moments queries. This is the scientific bridge from generic long-video QA to personal event recall.

The Ego4D mental model is fundamental:

The answer is not simply a sentence. It is a temporally grounded segment of the user's past visual experience.

For VoxSight, evaluation should preserve that grounding. Measure whether the correct event was retrieved before judging the quality of the spoken response. Then separately measure whether the response helps a user find or act on the object.

5.2 Personalized long-context retrieval

EgoMemory is a recent frontier paper on personalized retrieval from long-context egocentric video.¹⁵ It uses a personalized setting and highlights that user intent may require more than matching an object word to a caption. It reinforces a major research reality: **using a vector database over past observations is no longer novel by itself.**

The remaining opportunity is to build the retrieval and response layer around the demands of assistance: evidence provenance, uncertainty, state change, privacy, and accessibility. A model that finds a relevant video clip but tells a blind user an overconfident location is not a successful assistive system.

6. The closest system paper: MemPal

MemPal is the most important direct comparator for VoxSight Recall. It is a wearable, voice-based multimodal memory assistant for object retrieval in homes, designed with older adults.¹⁶ It uses selective visual capture rather than permanent raw-video storage: low-quality and redundant frames are removed; hand interactions can trigger potential placement events; a calibration video can help build a room map; and selected observations become text-based activity descriptions stored with object, room, time, and metadata.

MemPal establishes that the core use case—voice-based retrieval of a misplaced object—is both meaningful and technically feasible. It also sets a novelty boundary. A research paper cannot credibly claim that “a wearable camera plus a vector store can find misplaced items” is new.

What MemPal already demonstrates	What a differentiated VoxSight Recall paper could investigate
Wearable visual diary for voice object retrieval.	Provenance-first answers: source evidence, timestamp, and observation cues.
Event-oriented capture using hand interaction and room information.	State-aware memory: detect competing evidence of placement, pickup, move, or uncertainty.
An in-home use case with older adults.	Blind/low-vision interaction: concise spoken evidence, orientation cues, and actionability.
Selected observation and vector retrieval.	Confidence calibration: do not imply the object is still present after the last sighting.
Privacy-conscious reduction of raw media.	User-controllable memory: pause, inspect, delete, and set retention policies.

This is why the research opportunity should be framed as **trustworthy assistive episodic visual memory**, rather than as “another long-context visual-memory model.”

7. The system you should hold in your head

A useful implementation-level model separates **working context**, **event memory**, and **semantic memory**.

Memory tier	Contents	Retention	Query role	Example
Working context	Recent frames and dialogue.	Seconds/minutes .	Resolve what is happening now.	“The user is holding a keyring.”
Event memory	Time-stamped, evidence-linked observations.	Days/weeks, user-controlled.	Answer “when/where did I last...”	“10:18, keyring placed on kitchen counter.”
Semantic memory	Stable preferences,	Longer-lived, editable.	Interpret user terms and	“The user calls the side table

	room labels, object aliases.		personalize retrieval.	‘the desk.’ ”
Raw visual evidence	Keyframes/crops/ video clips, optionally encrypted.	Short or policy- controlled.	Verify retrieved event and enable review.	Keyframe showing object next to fruit bowl.

The runtime loop is then:

1. **Observe:** sample or stream camera frames and sensors.
2. **Detect candidate events:** use changes, hands, objects, user speech, or explicit “remember this” commands.
3. **Represent:** create compact event records with time, location cues, action/state, text and visual embeddings, and evidence links.
4. **Consolidate:** deduplicate near-identical events, merge confirmations, and apply retention/privacy policy.
5. **Retrieve:** convert a spoken request into an object/event query, then combine text, visual, temporal, and spatial scores.
6. **Verify:** examine the strongest evidence and look for later contradictory events.
7. **Respond:** speak the **last verified observation**, source time, confidence, and a next action; never pretend to know the present state without evidence.

A safe answer is: “I last saw your blue keyring placed on the kitchen counter beside the fruit bowl at 10:18. I have not seen it since, so it may have been moved.” An unsafe answer is: “Your keys are on the kitchen counter.”

The difference between those answers is the research gap you can own.

8. Open questions worth building a paper around

8.1 Provenance and verifiability

Most RAG systems retrieve evidence but do not always surface it meaningfully to an end user. In visual memory, provenance should include at least the source time, an accessible scene cue, and an internally retrievable keyframe. A good research question is:

Can provenance-aware spoken answers reduce harmful overtrust while preserving speed and usefulness in object-retrieval tasks?

8.2 Last-seen versus current location

An object can be visible in a past frame yet later be picked up, moved, or occluded. Existing episodic-localization tasks often treat the past occurrence as the answer. Assistive use requires a richer response policy. A good hypothesis is:

Modeling post-placement contradictory evidence improves the accuracy and calibration of “where is it?” responses compared with last-occurrence retrieval.

8.3 Event selection under a privacy and compute budget

Continuous recording is expensive and intrusive; uniform sampling can miss important actions. A strong systems question is:

Can interaction-triggered event capture retain object-placement recall quality while storing fewer frames and exposing less private visual data than uniform capture?

8.4 Accessibility and actionability

A retrieved clip is not an outcome. A blind or low-vision user needs short, spatially useful spoken help. A human-centered question is:

What evidence, uncertainty wording, and spatial cues allow a voice interface to turn visual recall into successful object recovery?

8.5 Evaluation beyond retrieval accuracy

Your evaluation hierarchy should be:

Level	Question	Example metric
Event retrieval	Did the system retrieve the correct past event?	Recall@K, temporal IoU, rank of ground-truth event.
State reasoning	Did it recognize later evidence of relocation or uncertainty?	Contradiction-detection precision/recall; calibration error.
Response faithfulness	Is the spoken answer supported by retrieved evidence?	Human/expert evidence-faithfulness rating.
Practical utility	Can a participant find the object or make the correct next move?	Task success, time-to-find, number of corrective turns.
Human factors	Does the user understand uncertainty and trust it	Comprehension, trust calibration, workload, privacy

appropriately?

acceptability.

9. Your prioritized reading sequence

Week 1 — Build the vocabulary and the historical argument

Read Bush first, then Tulving, then the SenseCam and MyLifeBits work. The goal is not to imitate their systems; it is to understand that personal memory technologies began as **human augmentation**, not as benchmark chasing. At the end of the week, write one page answering: “What is the difference between a stored visual event, a semantic fact, and a current-world claim?”

Priority	Read	Output you should produce
1	Bush (1945) 1	A three-sentence explanation of personal associative retrieval.
2	Tulving (1972) and Greenberg & Verfaellie (2010) 2 3	A precise operational definition of an event memory.
3	SenseCam (2006) and MyLifeBits (2006) 5 6	A table of capture, storage, retrieval, and user-control requirements.

Week 2 — Learn memory architectures without overcommitting to training

Read NTM to understand the memory abstraction, then RAG, Generative Agents, and MemGPT to understand modern retrieval and memory management. Your deliverable is an event-record schema and a retrieval-policy sketch. Do not train a model in this week.

Priority	Read	Question to answer
4	Neural Turing Machines 7	What is read/write memory conceptually?
5	RAG 8	What evidence should be retrieved before language is generated?

6	Generative Agents 9	Which observations should become durable memories?
7	MemGPT and MemoryBank 10 11	What is working memory versus long-term memory, and when should memory be forgotten?

Week 3 — Learn the visual-memory technical frontier

Read MovieChat and MA-LMM to understand long-video memory compression. Then study Ego4D. Write a comparison table separating **memory compression**, **event retrieval**, **state reasoning**, and **user communication**. This table will become part of a related-work section later.

Priority	Read	Question to answer
8	MovieChat 12	How can visual history be compressed, and what can be lost?
9	MA-LMM 13	Why process online rather than load all video into context?
10	Ego4D Episodic Memory 14	What is the benchmark's unit of ground truth and evaluation?
11	EgoMemory 15	What does personalization add beyond general video retrieval?

Week 4 — Study the closest application and define your gap

Read MemPal closely. Reproduce its system diagram in your notes, then list every component that VoxSight would retain, replace, or extend. Your final deliverable is a one-page research claim with a baseline, hypothesis, and evaluation plan.

Priority	Read	Question to answer
12	MemPal 16	What has already been demonstrated?

13	Ego4D privacy guidance ¹⁷	What privacy constraints should shape the architecture, not merely the ethics section?
14	Your own project code and this roadmap	What is currently absent in VoxSight: visual events, retrieval, provenance, state modeling, or user interaction?

10. Compact annotated bibliography

ID	Paper / source	Why it is essential	Reading depth
¹	Bush, <i>As We May Think</i> (1945)	Historical root of personal associative retrieval and the Memex.	Read carefully.
²	Tulving, <i>Episodic and Semantic Memory</i> (1972)	Foundational distinction for your terminology.	Read the conceptual sections / a reliable overview alongside it.
³	Greenberg & Verfaellie (2010)	Clarifies episodic–semantic relations.	Read carefully.
⁴	Boyle & Blomkvist (2024)	Prevents anthropomorphic misuse of “episodic memory” in AI.	Read carefully.
⁵	Gemmell et al., <i>MyLifeBits</i> (2006)	Personal database and lifelogging architecture.	Skim for system vision.
⁶	Hodges et al., <i>SenseCam</i> (2006)	Wearable retrospective memory aid.	Read carefully.
⁷	Graves et al., <i>Neural Turing Machines</i> (2014)	Neural external-memory abstraction.	Read abstract, figure, and method overview.
⁸	Lewis et al., <i>RAG</i> (2020)	Retrieval-plus-generation baseline and provenance motivation.	Read carefully.

9	Park et al., <i>Generative Agents</i> (2023)	Observation/memory/retrieval/reflection architecture.	Read carefully.
10	Packer et al., <i>MemGPT</i> (2023)	Tiered memory management for limited context.	Read overview and architecture.
11	Zhong et al., <i>MemoryBank</i> (2024)	Update/forgetting policy for long-term LLM memory.	Skim, then revisit if needed.
12	Song et al., <i>MovieChat</i> (2024)	Sparse memory for long-video understanding.	Read method and ablations.
13	He et al., <i>MA-LMM</i> (2024)	Online visual processing and long-term memory bank.	Read carefully.
14	Ego4D Episodic Memory benchmark	Task definitions, data, and evaluation.	Read documentation plus associated papers.
15	Tang et al., <i>EgoMemory</i> (2026)	Personalized long-context retrieval frontier.	Read carefully.
16	Maniar et al., <i>MemPal</i> (2025)	Closest wearable, voice-based object-retrieval comparator.	Read very carefully.
17	Ego4D Privacy and Ethics	Design constraints for first-person visual data.	Read before collecting any new data.

11. What not to confuse

Tempting statement	Why it is inaccurate	Better statement
“We are building a long-context model.”	You do not need to train a new foundation model to create useful persistent visual memory.	“We are building a retrieval- and event-memory layer over selected visual observations.”

“The model knows where the keys are.”	Past evidence does not establish present location.	“The system can report the last verified observation and explain its uncertainty.”
“Vector search is our contribution.”	This is a standard technique and already appears in related systems.	“Our contribution is evidence-aware, state-aware, accessible episodic retrieval.”
“Episodic memory means the AI remembers like a person.”	This overstates the analogy and makes claims hard to defend.	“The system stores and retrieves temporally grounded event records.”
“We should record everything.”	Continuous recording creates privacy, storage, and analysis burdens.	“We should selectively store high-value, user-controlled events.”

12. The recommended intellectual position for VoxSight

Your project should occupy this sentence:

VoxSight Recall studies how a privacy-conscious wearable assistant can convert selected first-person observations into evidence-grounded event memory, then answer object-location questions with accessible, calibrated spoken guidance.

This position is stronger than “large language visual memory,” more specific than generic long-video QA, and differentiated from existing object-retrieval systems by its focus on **provenance, uncertainty, state change, and accessibility**.

The next practical step after this reading is not a full implementation. It is to write a two-page problem statement defining: (1) the event schema, (2) the safe answer policy, (3) the first benchmark task, and (4) the one research hypothesis you will test.

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